

# NT219- Cryptography

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## Week 3: Modern Symmetric Ciphers

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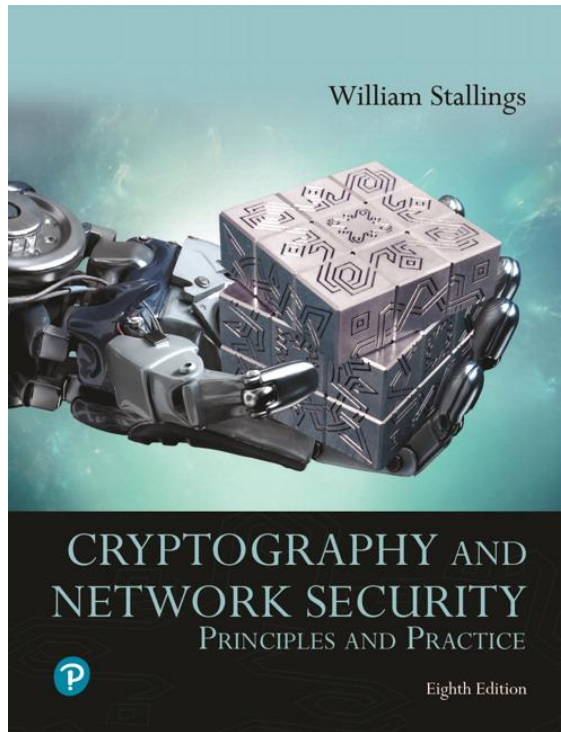
# Outline

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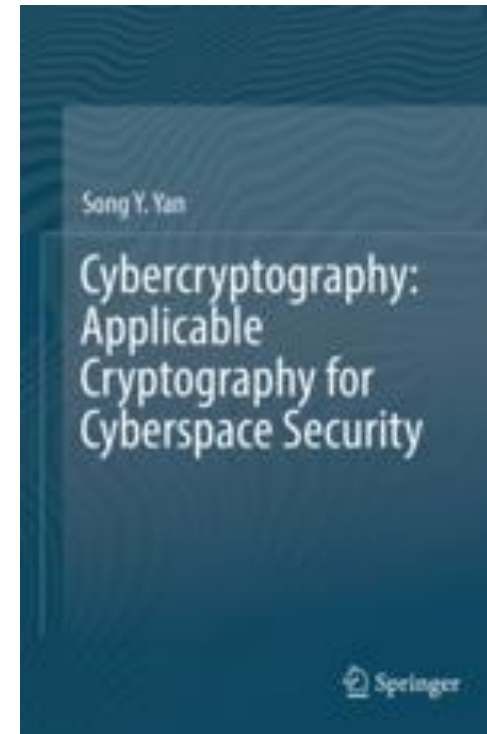
- Classical cipher algorithms (review)
- Stream Cipher
- Block cipher
  - Data Encryption Standard (DES)
  - Advanced Encryption Standard (AES)
  - Some other ciphers
    - Searchable encryption

# Textbooks and References

## ■ Text books



[1] Chapter 4,6



[2] Chapter 5

# Classical cipher algorithms

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## ■ Substitution Technique

### ➤ Monoalphabetic cipher

- Replace one character by another character

### ➤ Polyalphabetic cipher

- Replace some characters by other characters
  - 2 by 2:
  - 3 by 3:

## ■ Transposition Technique

- ### ➤ Keep the same source characters but change their positions

# Monoalphabetic cipher

## (1) Caesar Cipher (replace 1 character by 1 one character )

- Simplest and earliest known use of a substitution cipher
- Used by Julius Caesar

### Key

|        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plain  | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Cipher | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |

plain: MEET ME AFTER THE TOGA PARTY  
cipher: JBBQ JB XCQBO QEB QLDX MXOQV

# Monoalphabetic cipher

## (2) Generral Monoalphabetic substitution

|         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plain:  | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Cipher: | A | Z | E | R | T | Y | U | I | O | P | Q | S | D | F | G | H | J | K | L | M | W | X | C | V | B | N |

EX:

MEET ME AT OUR SPOT

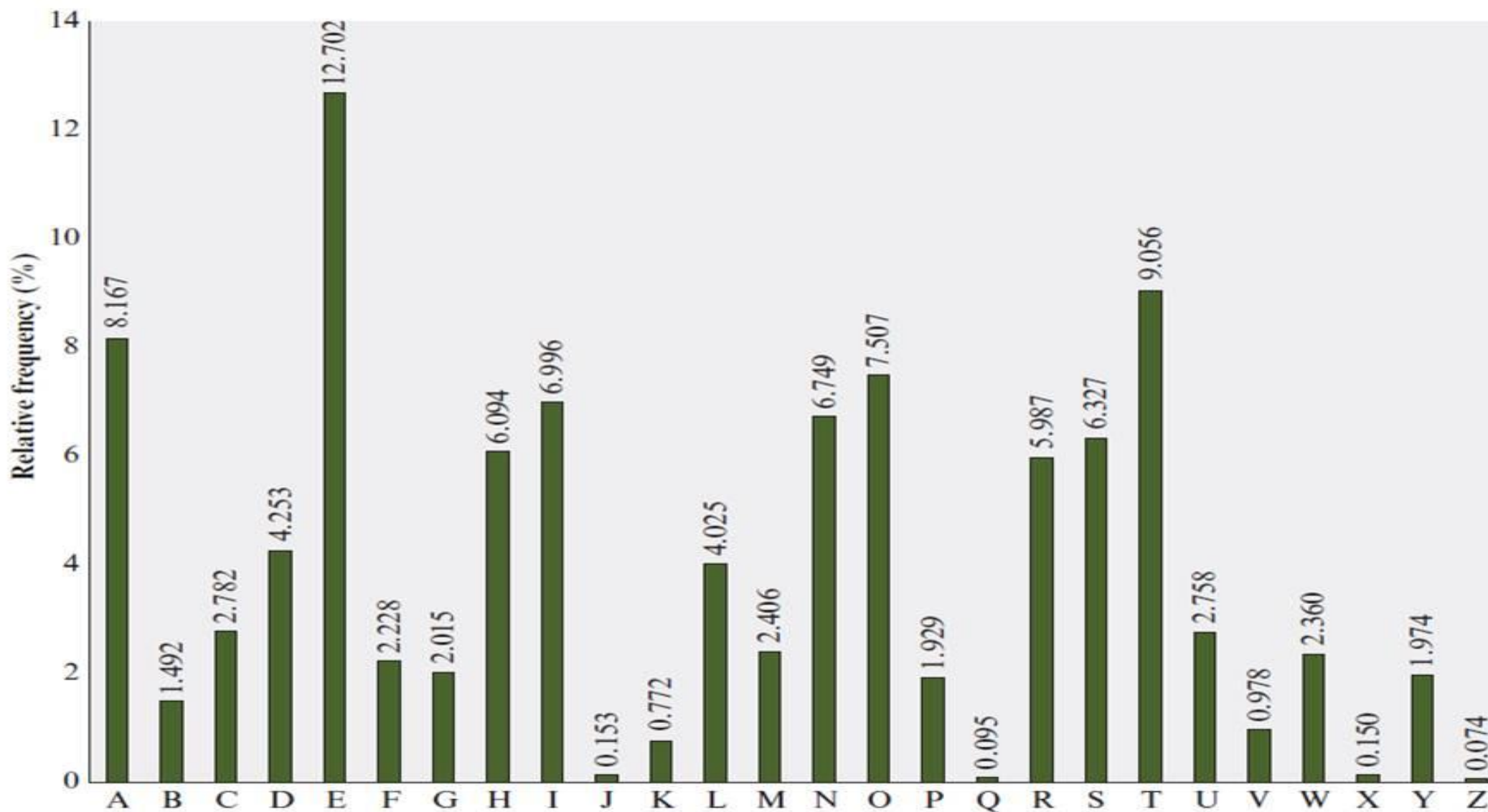


DTTM DT AM GWK LHGM

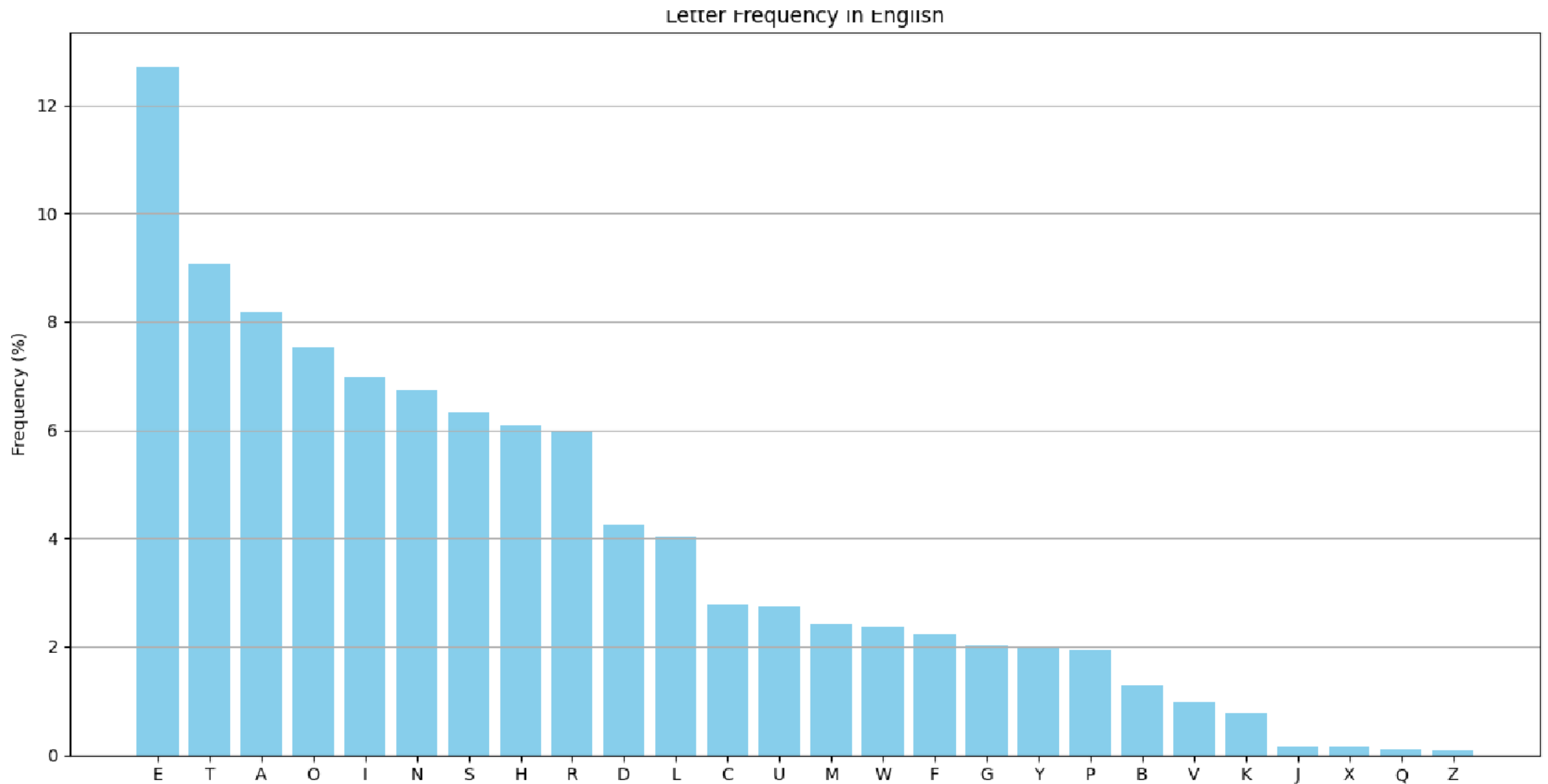
If the “cipher” line can be any permutation of the 26 alphabetic characters, then there are  $26!$  or greater than  $4 \times 10^{26}$  possible keys

This is 10 orders of magnitude greater than the key space for DES

# Relative Frequency of Letters in English Text



# Relative Frequency of Letters in English Text

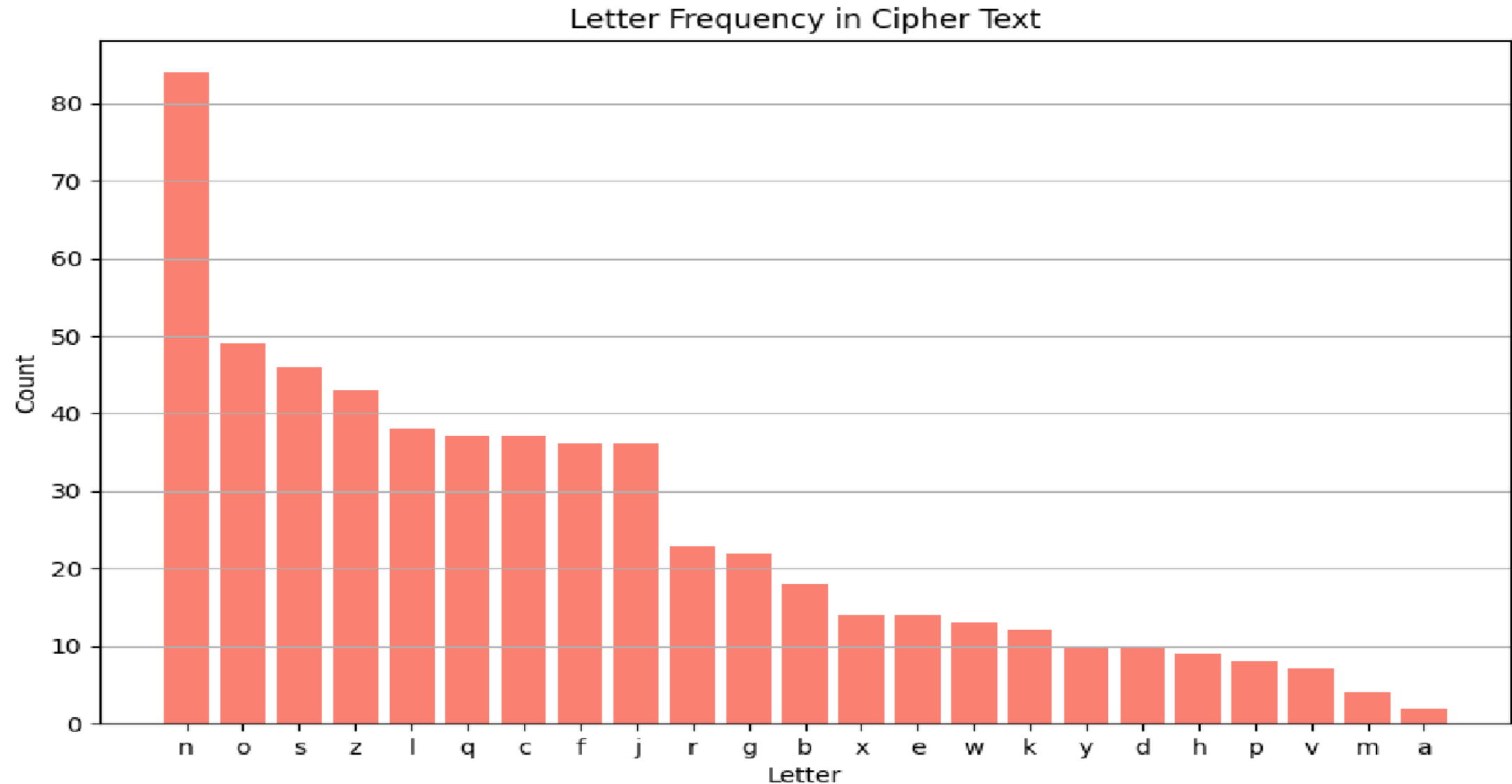


# Cryptanalysis on monoalphabetic cipher

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hzsrnqc klyy wqc flo mflwf ol zqdn nsoznj wskn lj xzsrbjnf, wzsxz gqv  
zqhhnf ol ozn glco zlfnc hnlhrn; nsoznj jnrqosdnc lj fnqj kjsnfbc, wzsxz  
sc xnjoqsfrv gljn efeceqr. zn rsdnb qrlfn sf zsc zlecn sf cqdsrrn jlw,  
wzsoznj flfn hnfnojqonb. q csfyrn blgncosx cekksxb ol cnjdn zsg. zn  
pjqmkqconb qfb bsfnb qo ozn xrep, qo zlejc gqozngqosxqrrv ksanb, sf  
ozn cqgn jllg, qo ozn cqgn oqprn, fndnj oqmsfy zsc gnqrc wsoz loznj  
gngpnjc, gexz rncc pjsfysfy q yenco wsoz zsg; qfb wnfo zlgn qo naqxorv  
gsbfsyzo, lfrv ol jnosjn qo lfxn ol pnb. zn fndnj ecnb ozn xlcx xzqgnjc  
wzsoznj ozn jnkljg hjldsbnc klj soc kqdlejnb gngpnjc. zn hqccnb onf zlejc  
leo lk ozn ownfov-klej sf cqdsrrn jlw, nsoznj sf crnnhsfy lj gqmsfy zsc  
olsrno.

# Cryptanalysis on monoalphabetic cipher



# Polyalphabetic Cipher

- Polyalphabetic Cipher is a **substitution** cipher in which the cipher alphabet for the plain alphabet may be **different at different places** during the encryption process;
  - Playfair Cipher: replace 2 characters by 2 characters
  - Hill Cipher: replace 3 characters by 3 characters
  - Vigenere Cipher



# Playfair encryption

Key matrix

|   |   |   |     |   |  |      |
|---|---|---|-----|---|--|------|
|   |   |   |     |   |  | → +1 |
| M | O | N | A   | R |  | ↓ +1 |
| C | H | Y | B   | D |  |      |
| E | F | G | I/J | K |  |      |
| L | P | Q | S   | T |  |      |
| U | V | W | X   | Z |  |      |

**Plaintext:** "Hide the gold in the tree stump"

**Plaintext diagram:**

HI DE TH EG OL DI NT HE TR EX ES TU MP

**Ciphertext diagram:**

BF CK PD FI .. .. .. ZD

[https://en.wikipedia.org/wiki/Playfair\\_cipher](https://en.wikipedia.org/wiki/Playfair_cipher)

# Polyalphabetic Cipher

## (4) Hill Cipher

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |

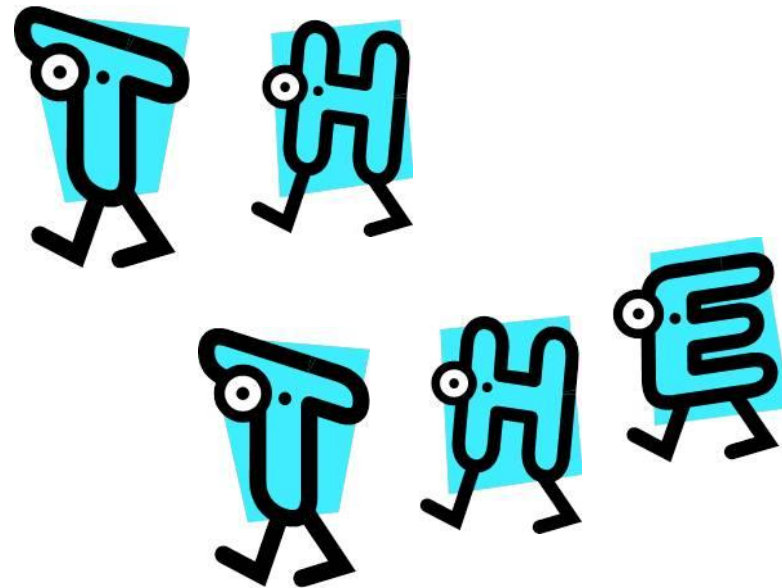
- Developed by the mathematician Lester Hill in 1929
- Strength is that it completely hides single-letter frequencies
  - The use of a larger matrix hides more frequency information
  - A 3 x 3 Hill cipher hides not only single-letter but also two-letter frequency information
- Strong against a ciphertext-only attack but easily broken with a known plaintext attack

$$C = K.P \bmod 26 \quad \begin{pmatrix} k_{1,1} & k_{1,2} & k_{1,3} \\ k_{2,1} & k_{2,2} & k_{2,3} \\ k_{3,1} & k_{3,2} & k_{3,3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \bmod 26$$

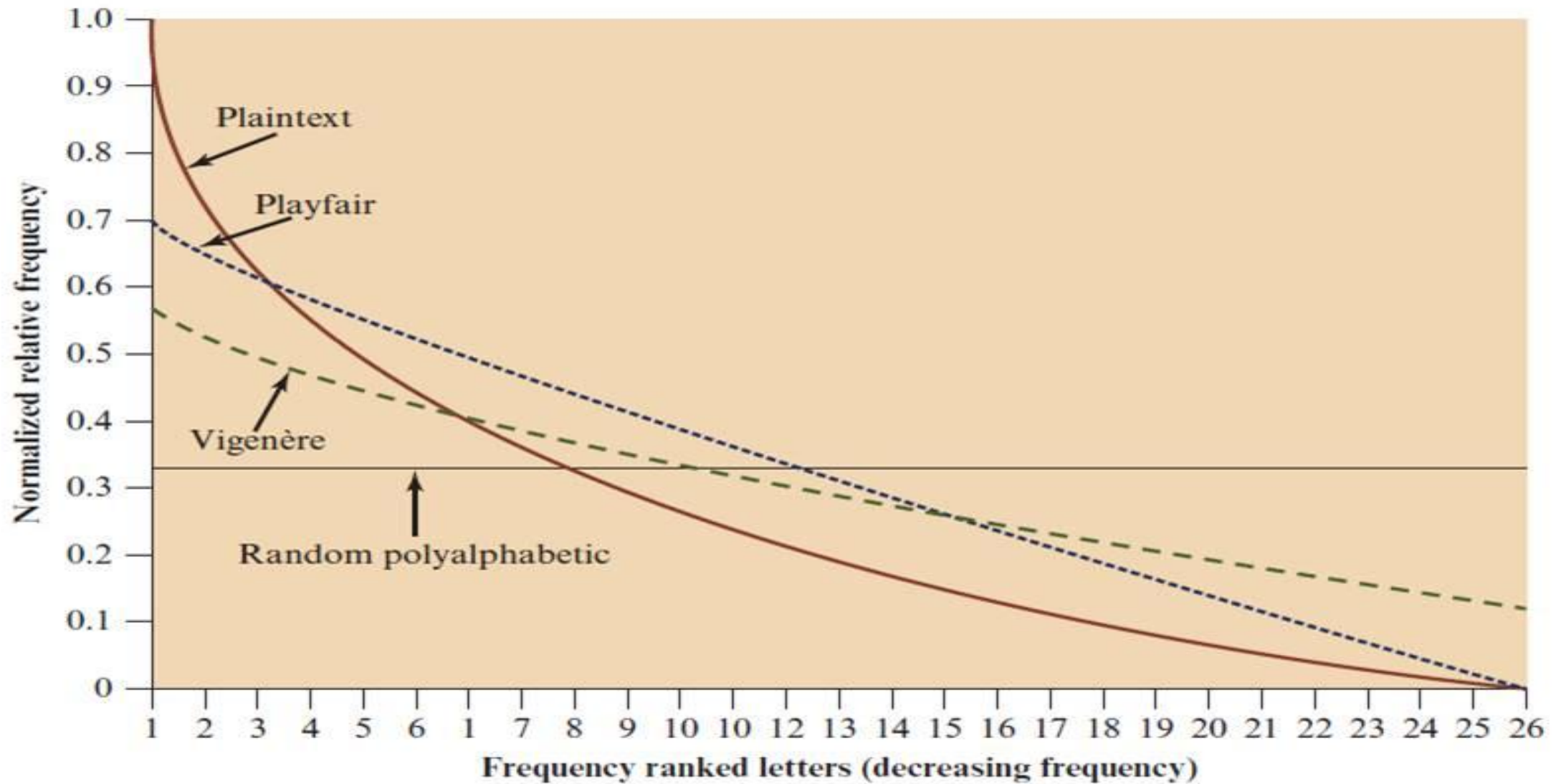
# Polyalphabetic Cipher

## Cryptoanalysis Playfair cipher

- Digram
  - Two-letter combination
  - Most common is *th*
- Trigram
  - Three-letter combination
  - Most frequent is *the*



# Classical symmetric cipher cryptanalysis



## (5) Vigenère Cipher

- Best known and one of the simplest polyalphabetic substitution ciphers
- In this scheme the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers with shifts of 0 through 25
- Each cipher is denoted by a key letter which is the ciphertext letter that substitutes for the plaintext letter a

[https://en.wikipedia.org/wiki/Vigen%C3%A8re\\_cipher](https://en.wikipedia.org/wiki/Vigen%C3%A8re_cipher)

# Example of Vigenère Cipher

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- To encrypt a message, a key is needed that is as long as the message
- Usually, the key is a repeating keyword
- For example, if the keyword is *deceptive*, the message “we are discovered save yourself” is encrypted as:

plaintext: wearediscoveredsaveyourself

key:       deceptivedeceptivedeceptive

ciphertext: ??

# Vigenère Cipher

## Vigenère matrix

|   | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

# Vigenère Autokey System

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- Example:

key:           deceptivewearediscoveredsav

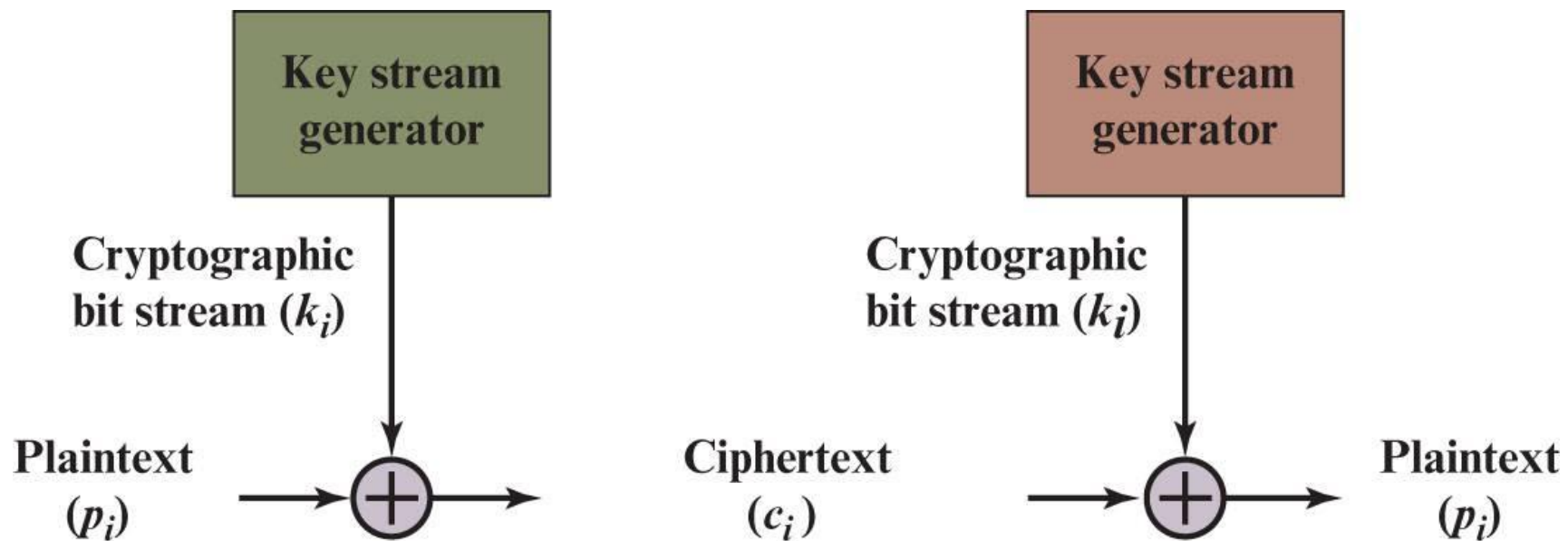
plaintext:   wearediscoveredsaveyourself

ciphertext:  ZICVTWQNGKZEIIGASXSTSLVWLA

- Even this scheme is vulnerable to cryptanalysis

- Because the key and the plaintext share the same frequency distribution of letters, a statistical technique can be applied

## (6) Vernam Cipher



[https://en.wikipedia.org/wiki/Gilbert\\_Vernam](https://en.wikipedia.org/wiki/Gilbert_Vernam)

# One-Time Pad

- Improvement to Vernam cipher proposed by an Army Signal Corp officer, Joseph Mauborgne
- Use a **random key that is as long as the message** so that the key need not be repeated
- Key is used to encrypt and decrypt a single message and then is discarded
- Each new message requires a new key of the same length as the new message
- **Scheme is unbreakable**
  - Produces random output that bears no statistical relationship to the plaintext
  - Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code



- The one-time pad offers complete security but, in practice, has two fundamental difficulties:
  - There is the practical problem of making large quantities of random keys
    - Any heavily used system might require millions of random characters on a regular basis
  - Mammoth key distribution problem
    - For every message to be sent, a key of equal length is needed by both sender and receiver
- Because of these difficulties, the one-time pad is of limited utility
  - Useful primarily for low-bandwidth channels requiring very high security
- The one-time pad is the only cryptosystem that exhibits *perfect secrecy* (see Appendix F)

# Transposition ciphers

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**Goals: scrambles the positions of characters**

**(1) Rail fence cipher**

**(2) Columnar Transposition Cipher**



[https://en.wikipedia.org/wiki/Transposition\\_cipher](https://en.wikipedia.org/wiki/Transposition_cipher)

# Transposition cipher

## (1) Rail fence cipher

- Simplest transposition cipher
- Plaintext is written down as a sequence of diagonals and then read off as a sequence of rows
- To encipher the message “meet me after the toga party” with a rail fence of depth 2, we would write:

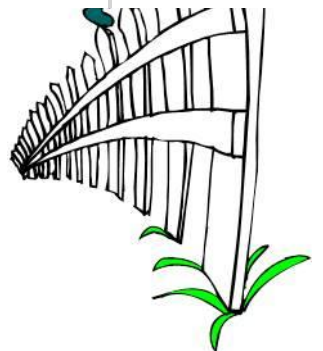
|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| m | e | m | a | t | r | h | t | g | p | r | y |
|   | e | t | e | f | e | t | e | o | a | a | t |

Ciphertext

Encrypted message is:

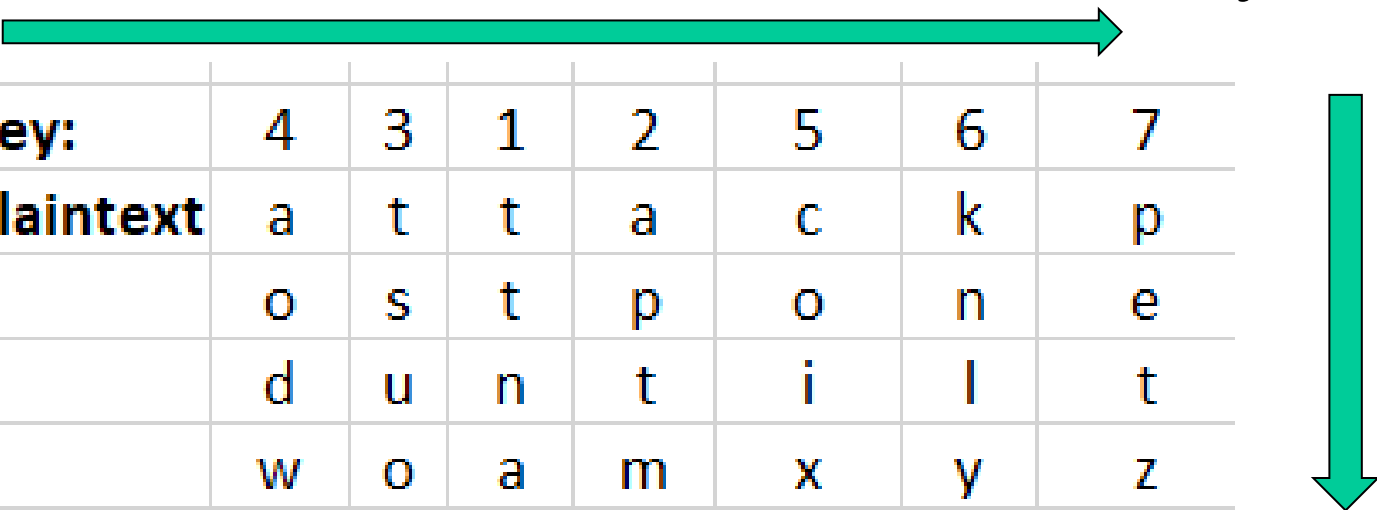
MEMATRHTGPRYETEFETEOAAT

[https://en.wikipedia.org/wiki/Rail\\_fence\\_cipher](https://en.wikipedia.org/wiki/Rail_fence_cipher)



# Columnar Transposition Cipher

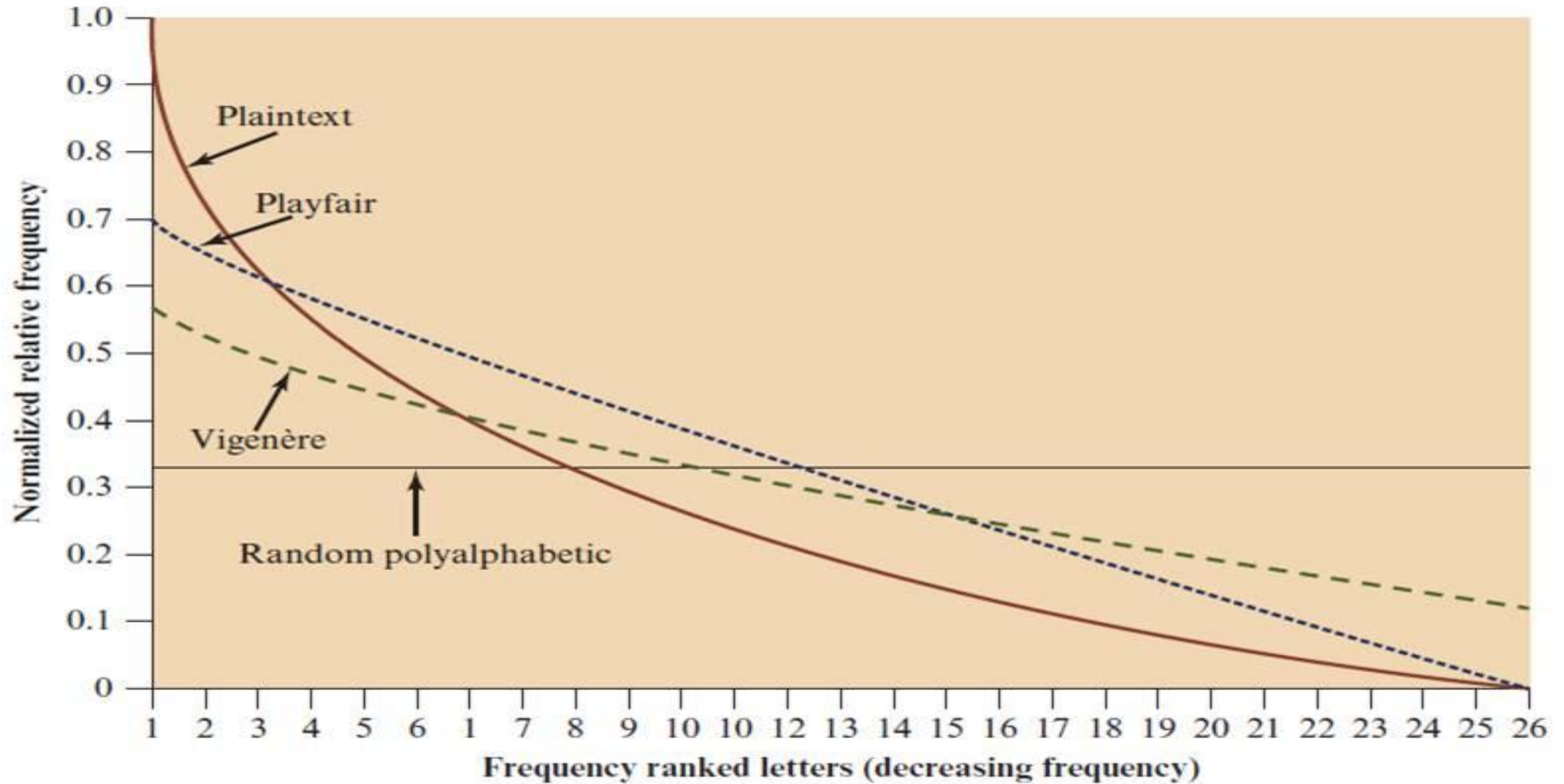
- Is a more complex transposition
- Write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns
  - The order of the columns then becomes the key to the algorithm



|                  |   |   |   |   |   |   |   |
|------------------|---|---|---|---|---|---|---|
| <b>Key:</b>      | 4 | 3 | 1 | 2 | 5 | 6 | 7 |
| <b>Plaintext</b> | a | t | t | a | c | k | p |
|                  | o | s | t | p | o | n | e |
|                  | d | u | n | t | i | l | t |
|                  | w | o | a | m | x | y | z |
|                  |   |   |   |   |   |   |   |

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

# Classical symmetric cipher cryptanalysis



# Stream Cipher (1 of 8)

- **Secret key (Keystream)**

$$K = k_1 k_2 \cdots k_i \cdots$$

- **Plaintext stream**

$$M = m_1 m_2 \cdots m_i \cdots$$

$m_i$  : bit or byte

- **Ciphertext**

$$C = c_1 c_2 \cdots c_i \cdots$$

where  $c_i = m_i \oplus \overline{k_i}$

$k_1 \quad k_2 \quad k_3 \quad \dots \quad k_n$

$m_1 \quad m_2 \quad m_3 \quad \dots \quad m_n$

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$k_1 \oplus m_1 \quad k_2 \oplus m_2 \quad \dots \quad k_n \oplus m_n$

# Stream Cipher (2 of 8)

## Vigenère cipher

| A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

### ■ Plaintext stream

|     |   |    |    |   |   |    |   |    |   |   |    |    |
|-----|---|----|----|---|---|----|---|----|---|---|----|----|
| M = | A | T  | T  | A | C | K  | A | T  | D | A | W  | N  |
|     | 0 | 19 | 19 | 0 | 2 | 10 | 0 | 19 | 3 | 0 | 22 | 13 |

### ■ Secret key (Keystream)

|      |    |   |    |    |    |    |   |    |    |    |    |   |
|------|----|---|----|----|----|----|---|----|----|----|----|---|
| K' = | L  | E | M  | O  | N  | L  | E | M  | O  | N  | L  | E |
|      | 11 | 4 | 12 | 14 | 13 | 11 | 4 | 12 | 14 | 13 | 11 | 4 |

### ➤ Ciphertext

|     |    |    |   |    |    |    |   |   |    |    |   |    |
|-----|----|----|---|----|----|----|---|---|----|----|---|----|
| C = | L  | X  | F | O  | P  | V  | E | F | R  | N  | H | R  |
|     | 11 | 23 | 5 | 14 | 15 | 21 | 4 | 5 | 17 | 13 | 7 | 17 |

$$C = c_1 c_2 \cdots c_i \cdots \text{ where } c_i = m_i + k_i \bmod 26$$

# Stream Cipher (3 of 8)

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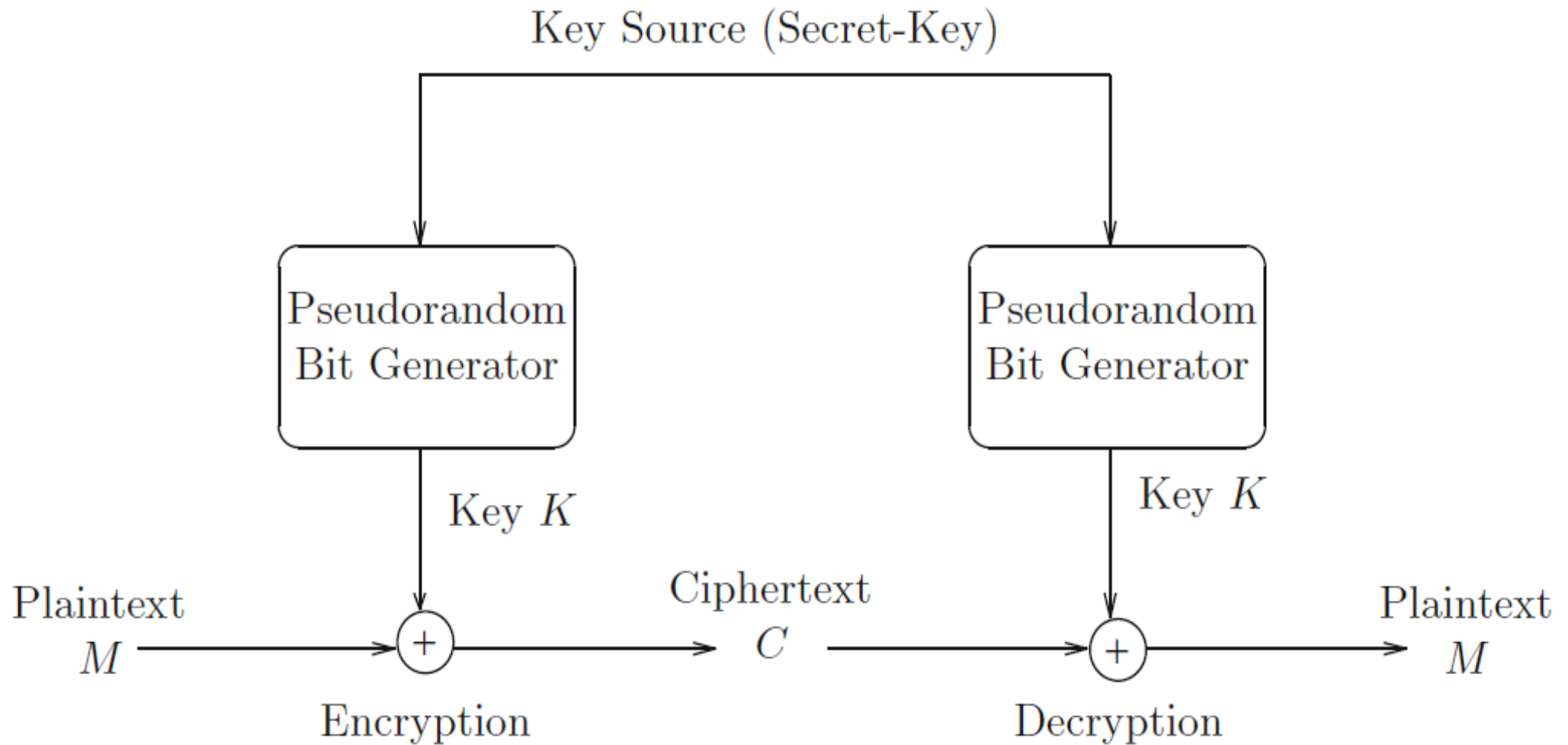
- Encrypts a digital data stream **one bit or one byte** at a time
  - Examples:
    - **Autokeyed** Vigenère cipher
    - Vernam cipher
- In the ideal case, a one-time pad version of the Vernam cipher would be used, in which the keystream is as long as the plaintext bit stream
  - If the cryptographic keystream is random, then this cipher is unbreakable by any means other than acquiring the keystream
    - Keystream must be provided to both users in advance via some independent and secure channel
    - This introduces insurmountable logistical problems if the intended data traffic is very large

# Stream Cipher (4 of 8)

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- For practical: must be implemented as an algorithmic to **generate key bit stream** (both users)
  - It must be computationally impractical to predict future portions of the bit stream based on previous portions of the bit stream
  - The two users need only share the **generating key** and each can produce the keystream

# Stream Cipher (5 of 8)



# Stream Cipher (6 of 8)

## ➤ Rivest Cipher 4

<https://en.wikipedia.org/wiki/RC4>

## ➤ Chaotic-based cryptosystem

[https://en.wikipedia.org/wiki/List\\_of\\_chaotic\\_maps](https://en.wikipedia.org/wiki/List_of_chaotic_maps)

| V · T · E           |                                                                                                                                                       | Stream ciphers                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Widely used ciphers | A5/1 · A5/2 · ChaCha · Crypto-1 · E0 · <b>RC4</b>                                                                                                     |                                       |
| eSTREAM Portfolio   | Software                                                                                                                                              | HC-256 · Rabbit · Salsa20 · SOSEMANUK |
|                     | Hardware                                                                                                                                              | Grain · MICKEY · Trivium              |
| Other ciphers       | Achterbahn · F-FCSR · FISH · ISAAC · MUGI · ORYX · Panama · Phelix · Pike · Py · QUAD · Scream · SEAL · SNOW · SOBER · SOBER-128 · VEST · VMPC · WAKE |                                       |
| Generators          | shrinking generator · self-shrinking generator · alternating step generator                                                                           |                                       |
| Theory              | block ciphers in stream mode · shift register · LFSR · NLFSR · T-function · IV                                                                        |                                       |
| Attacks             | correlation attack · correlation immunity · stream cipher attacks                                                                                     |                                       |

# Stream Cipher (7 of 8)

## ➤ Chaotic-based cryptosystem

Example:

Logistic map

$$x_{n+1} = rx_n(1 - x_n)$$

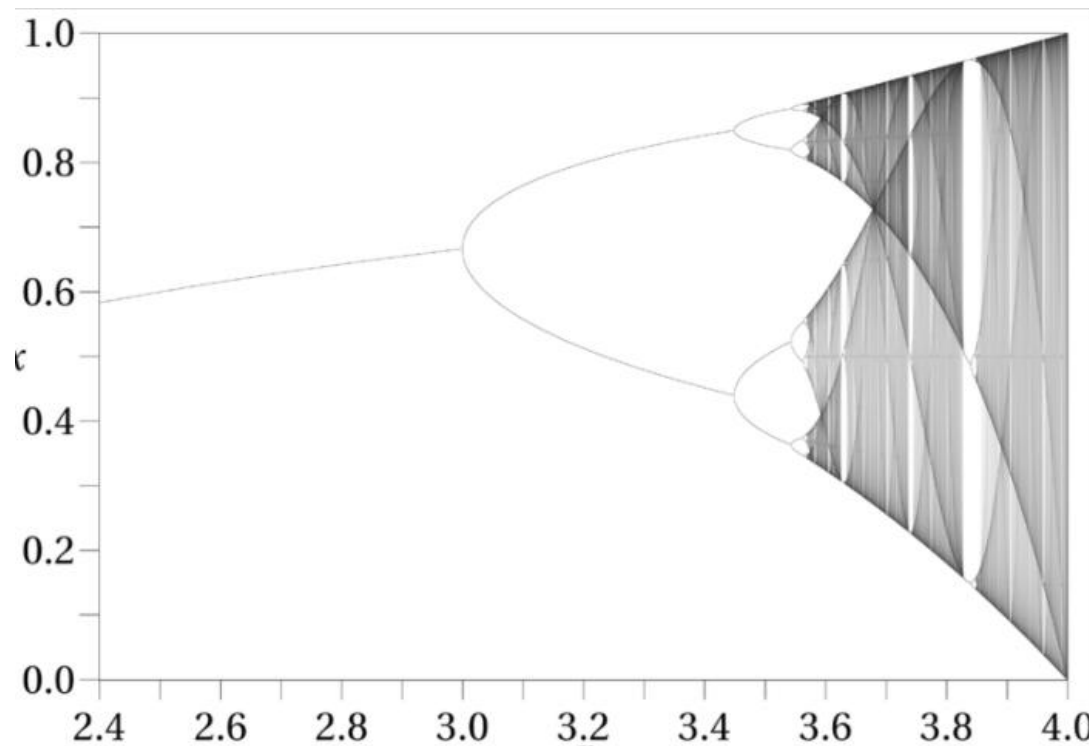
Input:

$$x_0, r \in (3.6, 4)$$

Output:

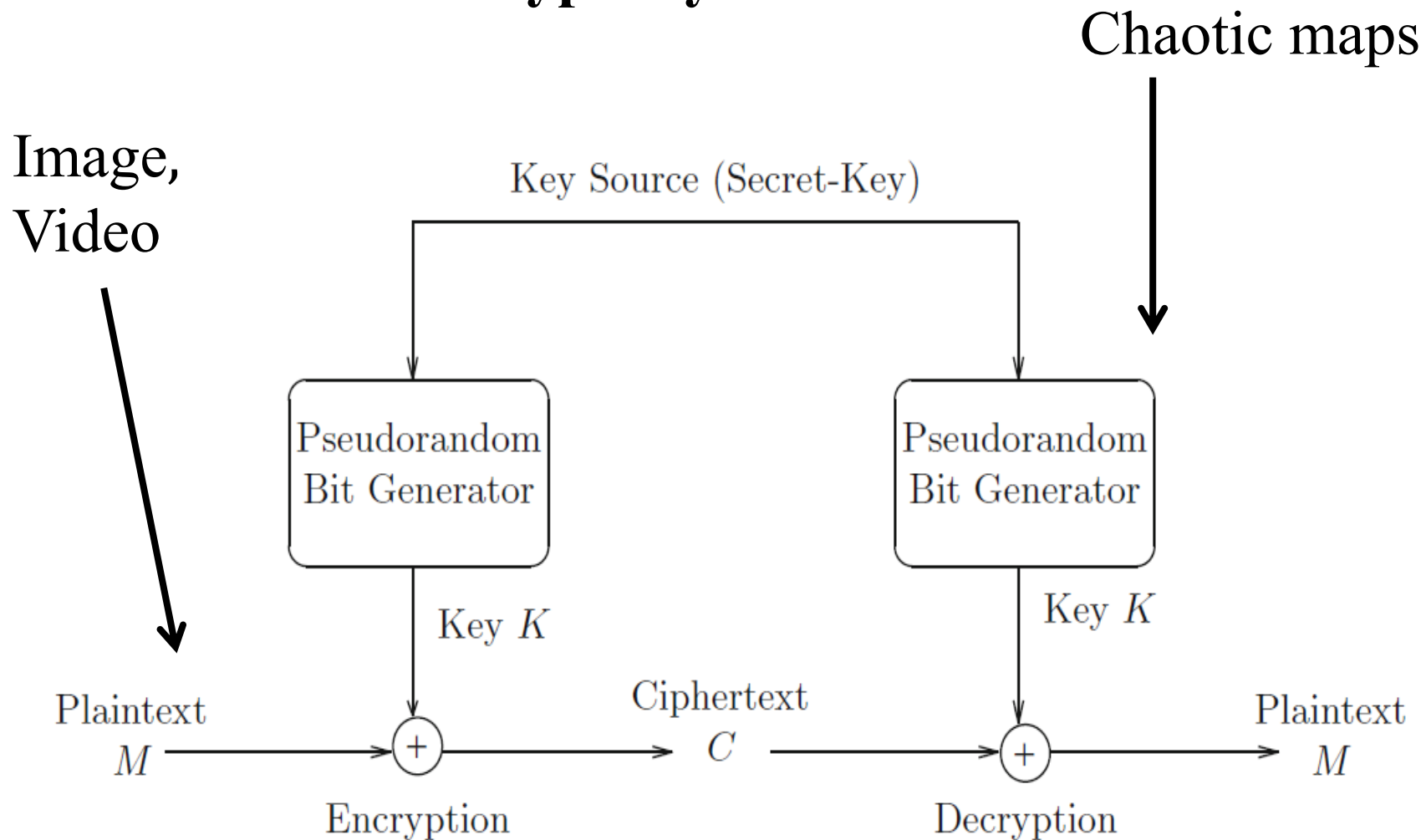
$$x_1, x_2, x_3, \dots, x_n, \dots$$

$$0 < x_i < 1$$



# Stream Cipher (8 of 8)

## ➤ Chaotic-based crypto system



# Outline

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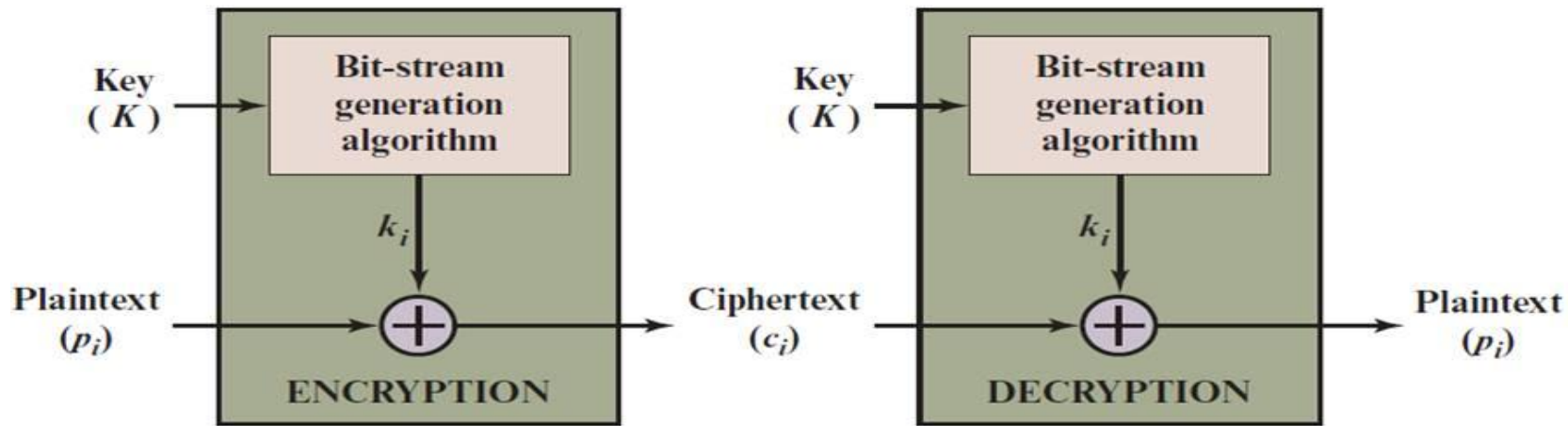
- Stream Cipher
- Block cipher
  - Data Encryption Standard (DES)
  - Advanced Encryption Standard (AES)
  - Some other ciphers
    - Searchable encryption

# Block Cipher

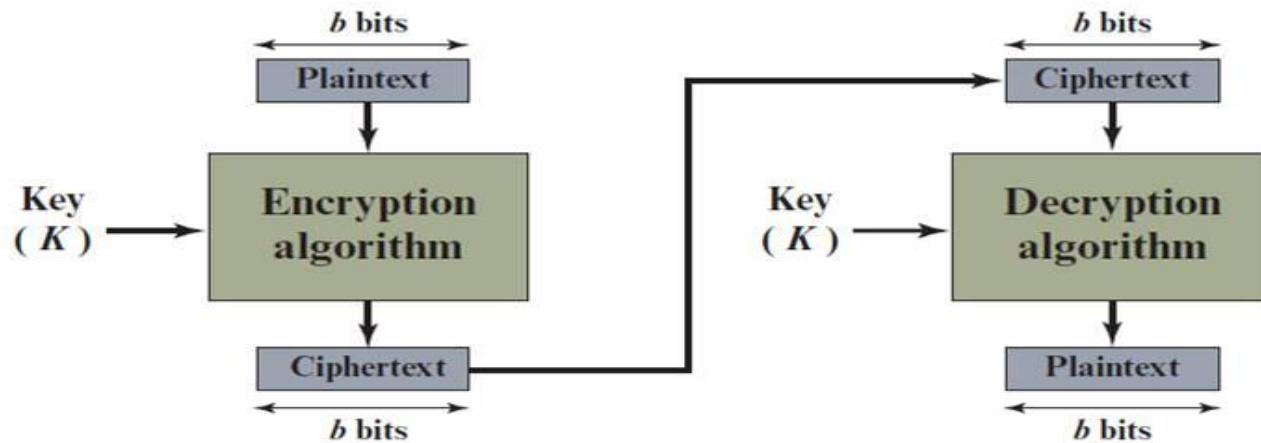
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- **A *block of plaintext* is treated as a whole** and used to produce a *ciphertext block of equal length*
- Typically a block size of 64 or 128 bits is used
- As with a stream cipher, the two users share a symmetric encryption key
- **The majority of network-based symmetric cryptographic applications make use of block ciphers**

# Stream Cipher Vs. Block Cipher



(a) Stream cipher using algorithmic bit-stream generator



(b) Block cipher

# Encryption and Decryption Tables for Substitution Cipher

$b_1 b_2 b_3 b_4$

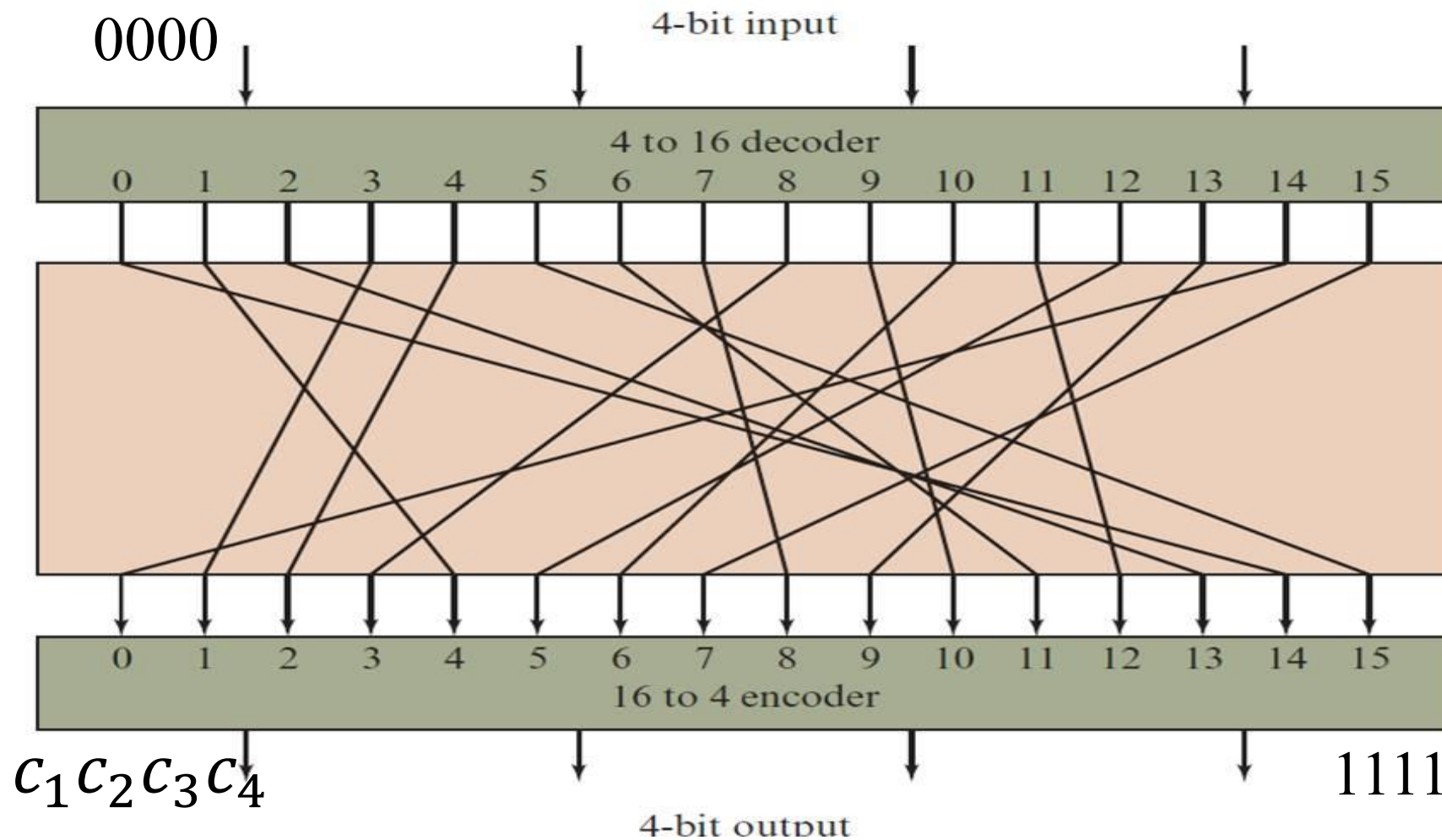
| Plaintext | Ciphertext |
|-----------|------------|
| 0000      | 1110       |
| 0001      | 0100       |
| 0010      | 1101       |
| 0011      | 0001       |
| 0100      | 0010       |
| 0101      | 1111       |
| 0110      | 1011       |
| 0111      | 1000       |
| 1000      | 0011       |
| 1001      | 1010       |
| 1010      | 0110       |
| 1011      | 1100       |
| 1100      | 0101       |
| 1101      | 1001       |
| 1110      | 0000       |
| 1111      | 0111       |

| Ciphertext | Plaintext |
|------------|-----------|
| 0000       | 1110      |
| 0001       | 0011      |
| 0010       | 0100      |
| 0011       | 1000      |
| 0100       | 0001      |
| 0101       | 1100      |
| 0110       | 1010      |
| 0111       | 1111      |
| 1000       | 0111      |
| 1001       | 1101      |
| 1010       | 1001      |
| 1011       | 0110      |
| 1100       | 1011      |
| 1101       | 0010      |
| 1110       | 0000      |
| 1111       | 0101      |

# Block Substitution

$b_1b_2b_3b_4$

## 4-bits block substitution



# Block Substitution

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How many possible substitutions for a block n-bit?

$$b_1 b_2 b_3 \dots b_n$$
$$c_1 c_2 c_3 \dots c_n$$

# Feistel Cipher

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- Feistel proposed the use of a cipher that alternates substitutions and permutations
- **Substitutions**
  - Each plaintext element or group of elements is uniquely **replaced** by a corresponding ciphertext element or group of elements
- **Permutation**
  - No elements are added or deleted or replaced in the sequence, rather the **order** in which the elements appear in the sequence is **changed**
- Is a practical application of a proposal by **Claude Shannon** to develop a product cipher that alternates **confusion and diffusion functions**
- Is the structure used by many significant symmetric block ciphers currently in use

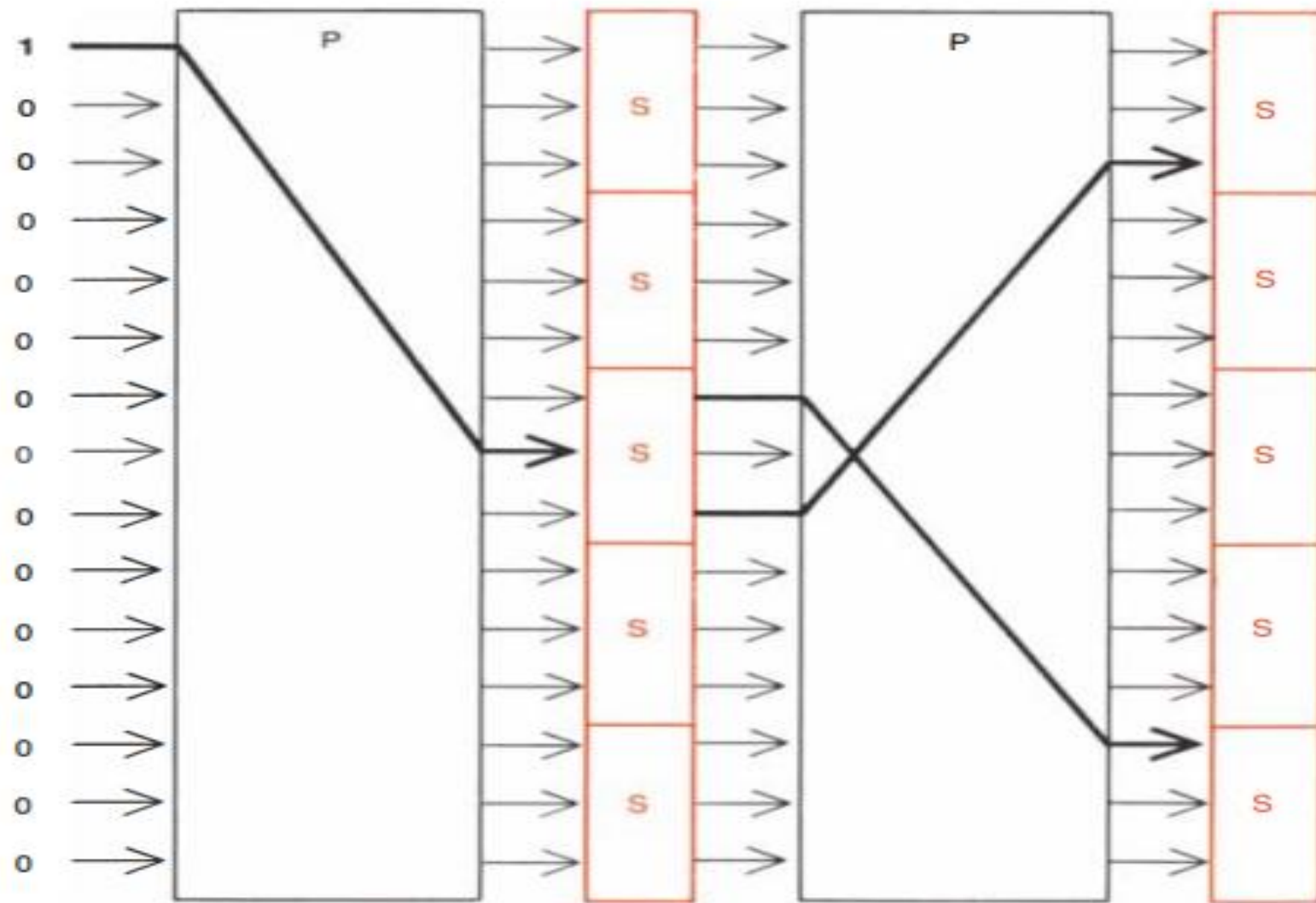
Feistel, H. (1973). Cryptography and computer privacy. Scientific american, 228(5), 15-23.

# Diffusion and Confusion

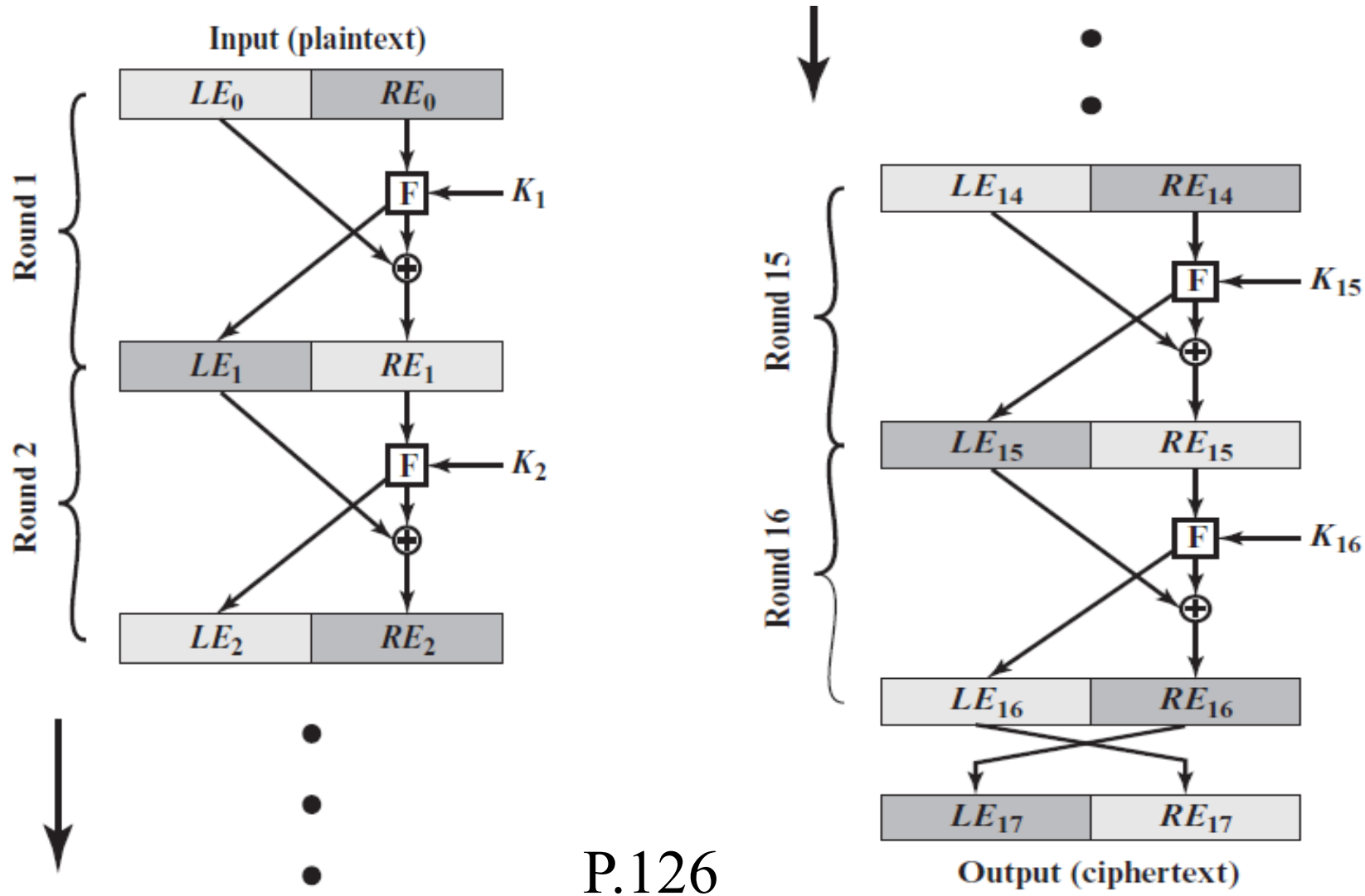
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- Terms introduced by Claude Shannon to capture the two basic building blocks for any cryptographic system
  - Shannon's concern was to thwart cryptanalysis based on statistical analysis
- **Diffusion**
  - The statistical structure of the **plaintext** is dissipated into long-range statistics of the **ciphertext**
  - This is achieved by having each plaintext digit affect the value of many ciphertext digits
- **Confusion**
  - Seeks to make the relationship between the statistics of the **ciphertext** and the value of the **encryption key** as complex as possible
  - Even if the attacker can get some handle on the statistics of the ciphertext, the way in which the key was used to produce that ciphertext is so complex as to make it difficult to deduce the key

# Feistel Cipher

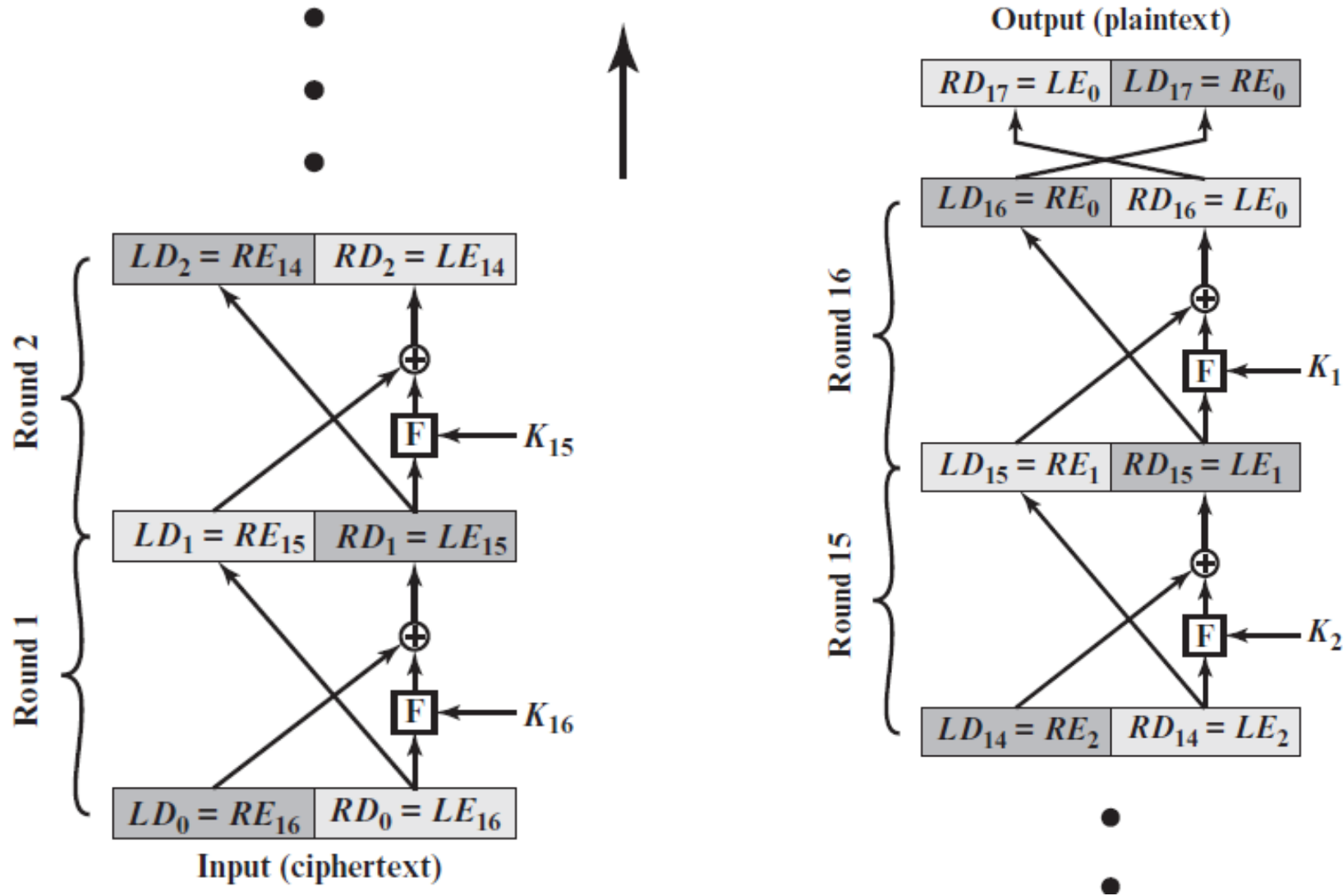


# Feistel Encryption and Decryption (16 rounds)



P.126

# Feistel Encryption and Decryption (16 rounds)



# The Feistel Cipher Scheme (FCS)

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- Divide  $M$  into blocks of  $2l$ -bits long (pad the last block if needed)
- Use only the **XOR** and **Substitution** operations
- Generate  $n$  **sub-keys** of a fixed length from the encryption key  $K$ :  $K_1, \dots, K_n$
- Divide a  $2l$ -bit block input into two parts:  $L_0$  and  $R_0$ , both of size  $l$  (the suffix and prefix of the block, respectively)
- Perform a substitution function  $F$  on an  $l$ -bit input string and a sub-key to produce an  $l$ -bit output
- Encryption and decryption each executes  $n$  rounds of the same sequence of operations

# FCS Encryption and Decryption

## FCS Encryption

- Let  $M = L_0R_0$ ; execute the following operations in round  $i, i = 1, \dots, n$ :

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

- Let  $L_{n+1} = R_n, R_{n+1} = L_n$  and  $C = L_{n+1}R_{n+1}$

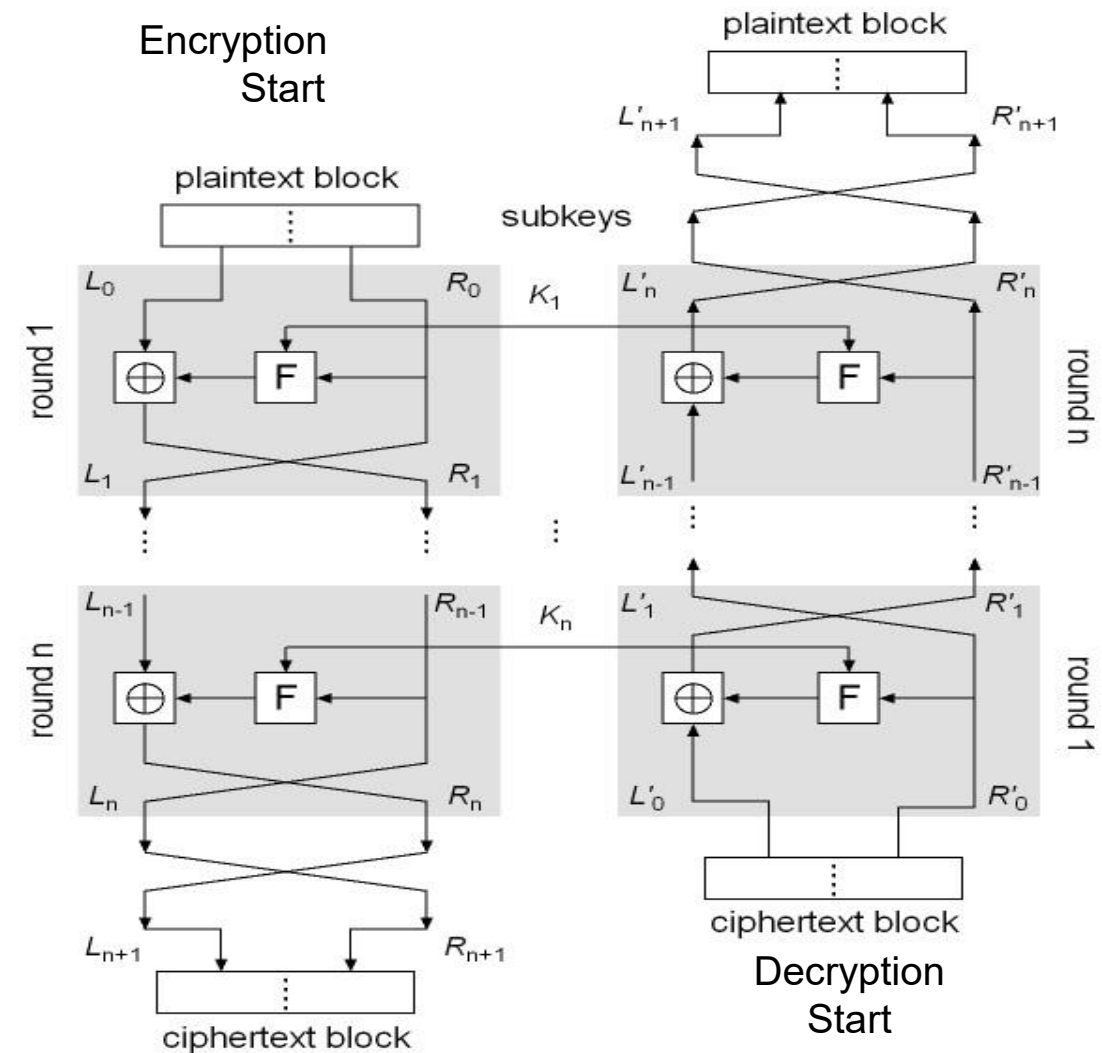
## FCS Decryption

- Symmetrical to encryption, with sub-keys in reverse order
- Rewrite  $C$  as  $C = L'_0R'_0$
- Execute the following in round  $i (i = 1, \dots, n)$ :

$$L'_i = R'_{i-1}$$

$$R'_i = L'_{i-1} \oplus F(R'_{i-1}, K'_{n-i+1})$$

- Let  $L'_{n+1} = R'_n, R'_{n+1} = L'_n$
- We will show that  $M = L'_{n+1}R'_{n+1}$



# Proof of FCS decryption

- Will show that  $C = L_{n+1}R_{n+1} = L'_0R'_0$  is transformed back to  $M = L_0R_0$  by the FCS Decryption algorithm
- Prove by induction the following equalities:

$$(1) L'_i = R_{n-i} \qquad (2) R'_i = L_{n-i}$$

- Basis:**  $L'_0 = L_{n+1} = R_n$ ,  $R'_0 = R_{n+1} = L_n$ ; (1) and (2) hold
- Hypothesis:** Assume when  $i \leq n$ :

$$L'_{i-1} = R_{n-(i-1)} \qquad R'_{i-1} = L_{n-(i-1)}$$

- Induction step:**

$$L'_i = R'_{i-1} \text{ (by decrypt. alg.)} = L_{n-i+1} \text{ (by hypothesis)} = R_{n-i} \text{ (by encrypt. alg.)}$$

Hence (1) is true

- $$\begin{aligned}
 R'_i &= L'_{i-1} \oplus F(R'_{i-1}, K_{n-i+1}) \\
 &= R_{n-(i+1)} \oplus F(L_{n-(i+1)}, K_{n-i+1}) \\
 &= [L_{n-i} \oplus F(R_{n-i}, K_{n-i+1})] \oplus F(R_{n-i}, K_{n-i+1}) \\
 &= L_{n-i}
 \end{aligned}$$

Hence (2) true

# Feistel Cipher Design Features (1 of 2)

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- Block size
  - Larger block sizes mean greater security but reduced encryption/decryption speed for a given algorithm
- Key size
  - Larger key size means greater security but may decrease encryption/decryption speeds
- Number of rounds
  - The essence of the Feistel cipher is that a single round offers inadequate security but that multiple rounds offer increasing security
- Subkey generation algorithm
  - Greater complexity in this algorithm should lead to greater difficulty of cryptanalysis

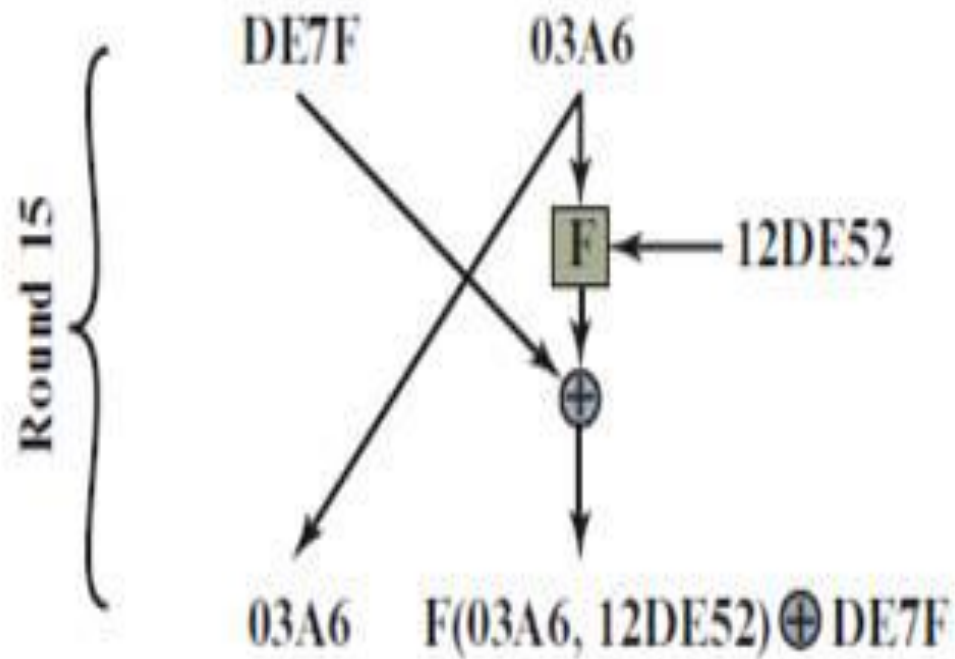
# Feistel Cipher Design Features (2 of 2)

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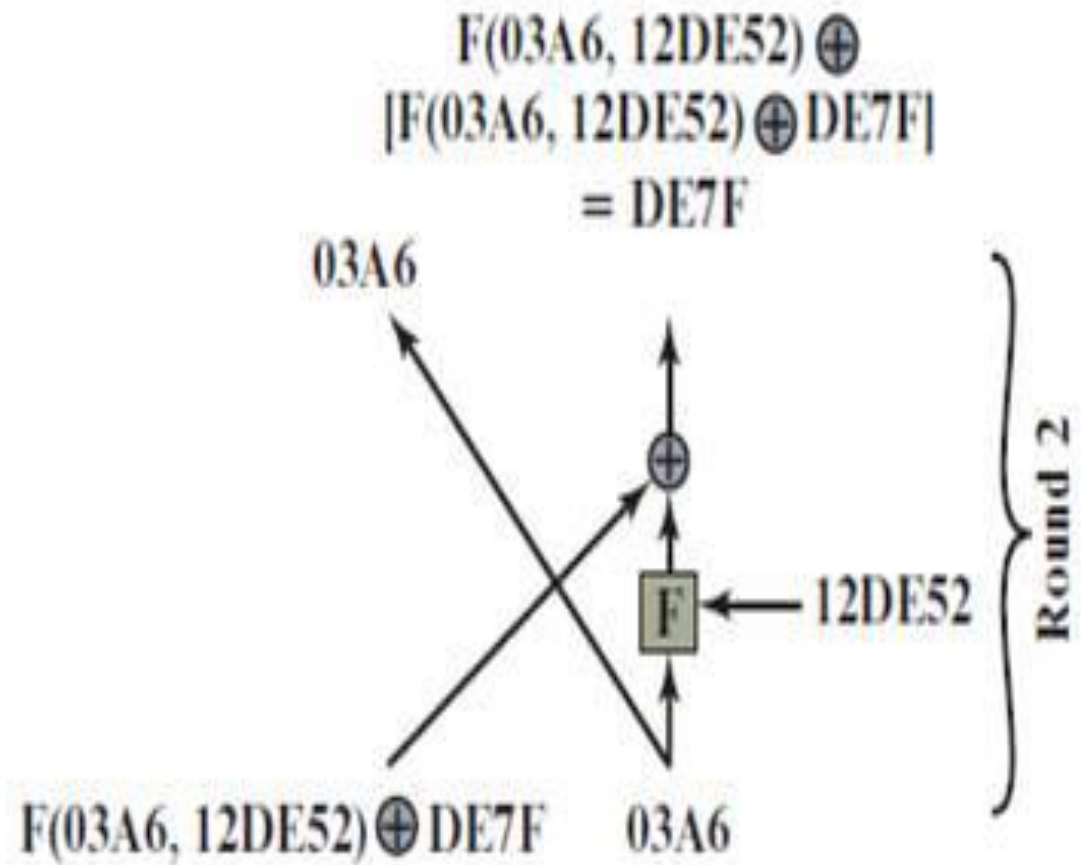
- Round function  $F$ 
  - Greater complexity generally means greater resistance to cryptanalysis
- Fast software encryption/decryption
  - In many cases, encrypting is embedded in applications or utility functions in such a way as to preclude a hardware implementation; accordingly, the speed of execution of the algorithm becomes a concern
- Ease of analysis
  - If the algorithm can be concisely and clearly explained, it is easier to analyze that algorithm for cryptanalytic vulnerabilities and therefore develop a higher level of assurance as to its strength

# Feistel Example

Encryption round



Decryption round



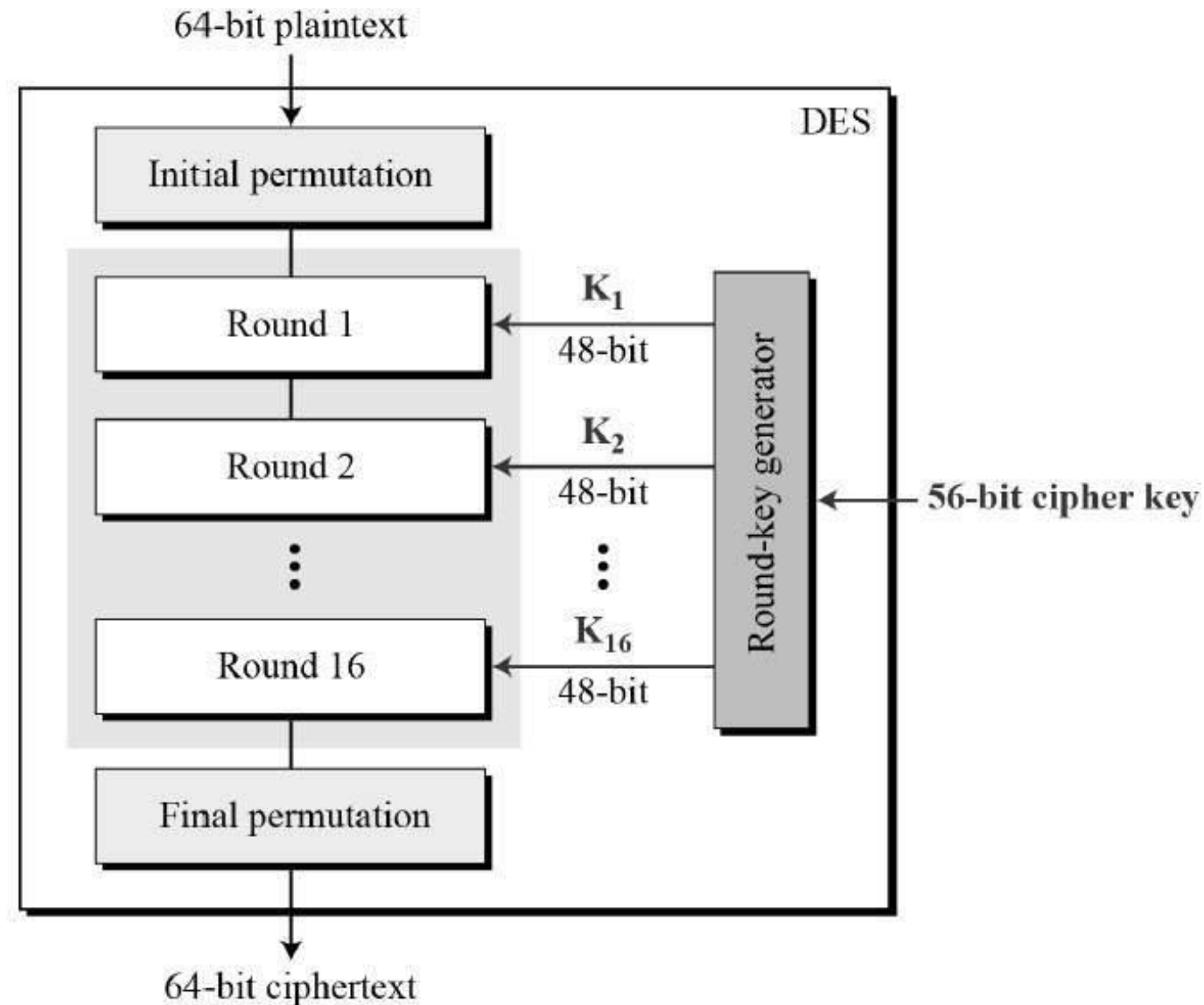
- Stream Cipher
- Block cipher
  - Data Encryption Standard (DES)
  - Advanced Encryption Standard (AES)
  - Some other ciphers
    - Searchable encryption

# Data Encryption Standard (DES)

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- Issued in 1977 by the National Bureau of Standards (now NIST) as Federal Information Processing Standard 46
- Was the most widely used encryption scheme until the introduction of the Advanced Encryption Standard (AES) in 2001
- Algorithm itself is referred to as the Data Encryption Algorithm (DEA)
  - Data are encrypted in 64-bit blocks using a 56-bit key
  - The algorithm transforms 64-bit input in a series of steps into a 64-bit output
  - The same steps, with the same key, are used to reverse the encryption

# DES Encryption Algorithm



# DES Sub-Key Generation

- The block size of DES is 64 bits and the encryption key is 56 bits, which is represented as a 64-bit string  $K = k_1 k_2 \dots k_{64}$
- DES uses 16 rounds of iterations with 16 sub-keys
- Sub-key generation:
  1. Remove the  $8i$ -th bit ( $i = 1, 2, \dots, 8$ ) from  $K$
  2. Perform an **initial permutation** on the remaining 56 bits of  $K$ , denoted by  $IP_{\text{key}}(K)$
  3. Split this 56-bit key into two pieces:  $U_0 V_0$ , both with 28 bits
  4. Perform Left Circular Shift on  $U_0$  and  $V_0$  a defined number of times, producing  $U_i V_i$ :
$$U_i = LS_{Z(i)}(U_{i-1}), \quad V_i = LS_{Z(i)}(V_{i-1})$$
  5. Permute the resulting  $U_i V_i$  using a defined compress permutation, resulting in a 48-bit string as a sub-key, denoted by  $K_i$ 
$$K_i = P_{\text{key}}(U_i V_i)$$

<https://www.geeksforgeeks.org/data-encryption-standard-des-set-1/>

# DES Substitution Boxes

- The DES substitution function  $F$  is defined below:

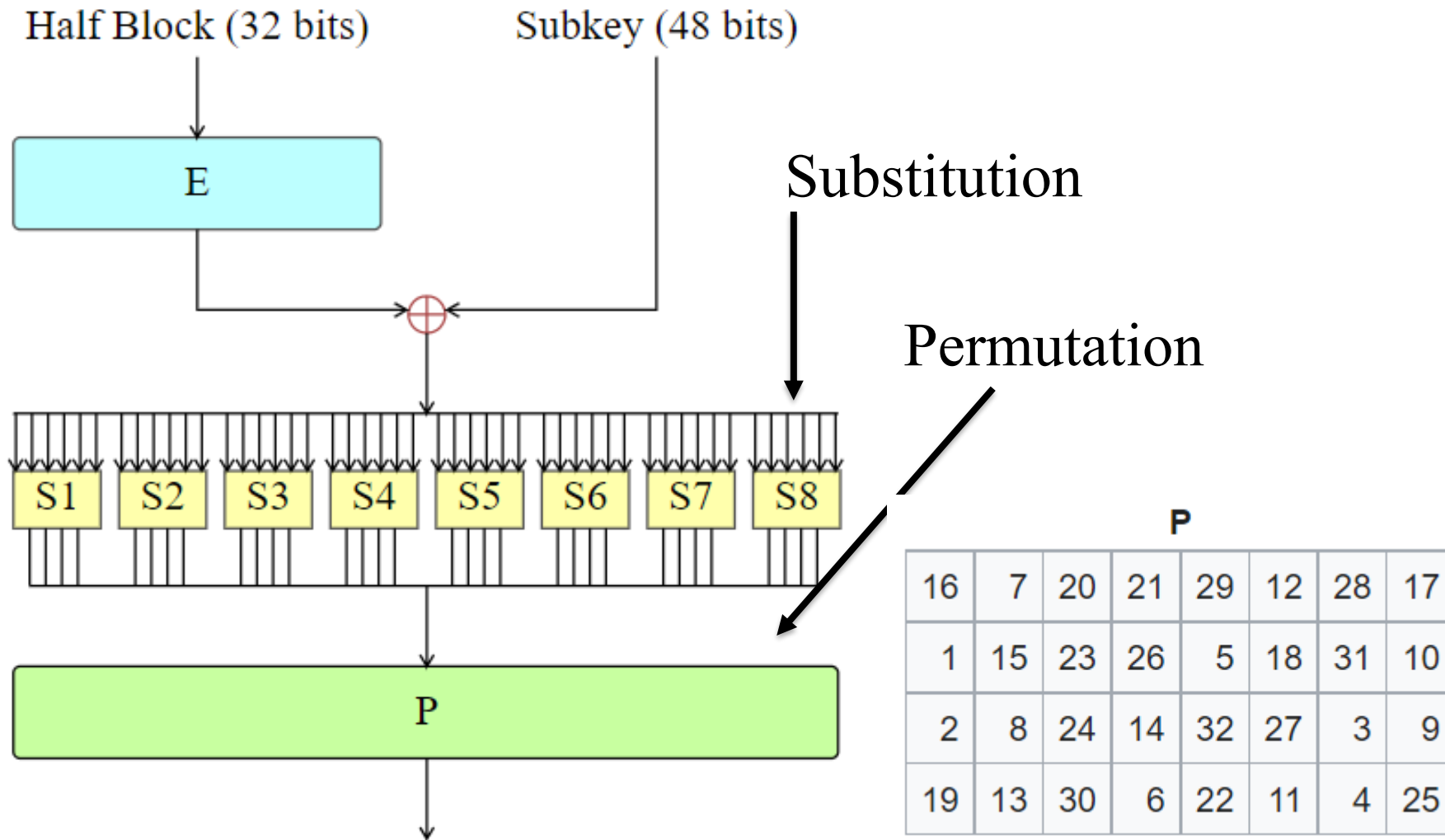
$$F(R_{i-1}, K_i) = P(S(EP(R_{i-1}) \oplus K_i)), i = 1, \dots, 16$$

- First, permute  $R_i$  using  $EP(R_i)$  to produce a 48-bit string  $x$
- Next, XOR  $x$  with the 48-bit sub key  $K_i$  to produce a 48-bit string  $y$
- Function  $S$  turns  $y$  into a 32-bits string  $z$ , using eight 4x16 special matrices, called S-boxes
  - Each entry in an S-box is a 4-bit string
  - Break  $y$  into 8 blocks, each with 6-bits
  - Use the  $i^{\text{th}}$  matrix on the  $i^{\text{th}}$  block  $b_1b_2b_3b_4b_5b_6$
  - Let  $b_1b_6$  be the row number, and  $b_2b_3b_4b_5$  the column number, and return the corresponding entry
  - Each 6-bit block is turned to a 4-bit string, resulting in a 32-bit string  $z$
- Finally, permute  $z$  using  $P$  to produce the result of DES's F function
- This result, XOR'd with  $L_{i-1}$ , is  $R_i$

[https://en.wikipedia.org/wiki/DES\\_supplementary\\_material](https://en.wikipedia.org/wiki/DES_supplementary_material)

# DES function $F(R_{i-1}, K_i)$

$$F(R_{i-1}, K_i) = P(S(EP(R_{i-1}) \oplus K_i)), i = 1, \dots, 16$$



# DES Substitution Boxes

| S <sub>5</sub> |    | Middle 4 bits of input |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----------------|----|------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|                |    | 0000                   | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
| Outer bits     | 00 | 0010                   | 1100 | 0100 | 0001 | 0111 | 1010 | 1011 | 0110 | 1000 | 0101 | 0011 | 1111 | 1101 | 0000 | 1110 | 1001 |
|                | 01 | 1110                   | 1011 | 0010 | 1100 | 0100 | 0111 | 1101 | 0001 | 0101 | 0000 | 1111 | 1010 | 0011 | 1001 | 1000 | 0110 |
|                | 10 | 0100                   | 0010 | 0001 | 1011 | 1010 | 1101 | 0111 | 1000 | 1111 | 1001 | 1100 | 0101 | 0110 | 0011 | 0000 | 1110 |
|                | 11 | 1011                   | 1000 | 1100 | 0111 | 0001 | 1110 | 0010 | 1101 | 0110 | 1111 | 0000 | 1001 | 1010 | 0100 | 0101 | 0011 |

Input: “**011011**”

Output: “1001”

# DES encryption steps

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- Rewrite  $IP(M) = L_0R_0$ , where  $|L_0| = |R_0| = 32$
- For  $i = 1, 2, \dots, 16$ , execute the following operations in order:  
$$L_i = R_{i-1}$$
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$
- Let  $C = IP^{-1}(R_{16}L_{16})$ .

# Is DES good enough?

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- Security strength of DES
  - Number of rounds
  - Length of encryption key
  - Construction of the substitute function
- DES was used up to the 1990's.
- People began to take on the DES Challenges to crack DES
- Only uses 56-bit keys =  $2^{56} \sim 7.2 \times 10^{16}$  keys
- Brute-force will work with current technology
  - In 1997 on Internet in a few months
  - In 1998 on dedicated h/w (EFF) in a few days
  - In 1999 above combined in 22 hours

# What to Do Next?

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- Start over
- New standards begin to be looked into
- On the other hand, can we extend the use of DES?

# Block Cipher Design Principles

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- The greater the number of rounds, the more difficult it is to perform cryptanalysis
- In general, the criterion should be that the number of rounds is chosen so that known cryptanalytic efforts require greater effort than a simple brute-force key search attack
- If DES had 15 or fewer rounds, differential cryptanalysis would require less effort than a brute-force key search

# Block Cipher Design Principles

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- The heart of a Feistel block cipher is the function  $F$
- The more nonlinear  $F$ , the more difficult any type of cryptanalysis will be
- The SAC and BIC criteria appear to strengthen the effectiveness of the confusion function

## **The algorithm should have good avalanche properties**

- Strict avalanche criterion (SAC)
  - States that any output bit  $j$  of an S-box should change with probability  $1/2$  when any single input bit  $i$  is inverted for all  $i, j$
- Bit independence criterion (BIC)
  - States that output bits  $j$  and  $k$  should change independently when any single input bit  $i$  is inverted for all  $i, j$ , and  $k$

# Block Cipher Design Principles

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- With any Feistel block cipher, the key is used to generate one subkey for each round
- In general, we would like to select subkeys to maximize the difficulty of deducing individual subkeys and the difficulty of working back to the main key
- It is suggested that, at a minimum, the key schedule should guarantee key/ciphertext Strict Avalanche Criterion and Bit Independence Criterion